

WAIOURU, New Zealand

WMO: 933390

Lat: 39.446S

Lon: 175.658E

Elev: 819

StdP: 91.87

Time Zone: 12.00 (NZL)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-3.9	-2.7	-5.8	2.6	-2.8	-4.4	2.9	-1.7	15.4	6.7	12.2	7.1	1.4	100	0.601

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	9.9	22.4	16.0	21.1	15.3	19.9	14.8	17.3	20.6	16.4	19.6	15.6	18.7	3.7	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.1	12.6	19.1	15.1	11.9	18.1	14.2	11.2	17.4	52.0	20.7	49.2	19.7	46.6	18.8	20.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.8	9.0	7.8	DB	-6.4	25.2	1.0	1.6	-7.0	26.4	-7.6	27.3	-8.1	28.1	-8.8	29.3	(4)
(5)			WB	-6.9	19.0	1.0	0.9	-7.6	19.6	-8.2	20.2	-8.7	20.7	-9.4	21.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	8.8	13.8	14.1	12.2	9.4	7.2	4.7	4.0	4.6	6.1	7.6	9.5	12.3	(6)
(7)		DBStd	4.43	2.92	2.72	2.80	2.83	2.99	2.86	2.52	2.42	2.58	2.66	2.84	2.77	(7)
(8)		HDD10.0	929	6	3	13	43	96	160	185	169	120	81	43	12	(8)
(9)		HDD18.3	3497	142	120	192	267	344	409	443	427	368	331	266	188	(9)
(10)		CDD10.0	476	123	117	79	26	9	1	0	0	2	8	27	82	(10)
(11)		CDD18.3	2	1	1	0	0	0	0	0	0	0	0	0	0	(11)
(12)		CDH23.3	18	10	6	1	0	0	0	0	0	0	0	0	1	(12)
(13)		CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	3.0	2.9	2.7	2.6	2.5	2.7	2.9	2.9	3.1	3.4	3.5	3.4	3.3	(14)
(15)	Precipitation	PrecAvg	1985	137	108	131	145	169	208	200	188	192	181	155	175	(15)
(16)		PrecMax	2403	225	453	252	251	288	318	354	290	468	384	245	259	(16)
(17)		PrecMin	1617	55	18	56	88	67	111	71	75	38	72	62	79	(17)
(18)		PrecStd	244	36	88	53	48	51	60	76	58	88	60	54	46	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.8	24.1	22.0	18.6	16.2	13.5	11.8	12.4	15.1	17.0	20.7	22.0	(19)
(20)			MCWB	16.3	17.0	14.5	13.8	12.6	10.9	9.3	8.9	10.9	12.0	14.0	15.2	(20)
(21)		2%	DB	22.6	22.4	20.2	16.8	14.5	11.5	10.3	10.9	13.3	15.3	18.4	20.2	(21)
(22)			MCWB	16.4	16.2	14.6	12.9	11.8	9.5	8.1	7.9	9.6	10.9	13.0	14.6	(22)
(23)		5%	DB	21.1	21.0	18.9	15.6	13.2	10.3	9.2	9.7	12.1	14.1	16.7	19.0	(23)
(24)			MCWB	15.5	15.6	14.0	12.5	10.9	8.5	7.2	7.2	9.1	10.2	12.2	14.2	(24)
(25)		10%	DB	19.8	19.7	17.6	14.5	12.0	9.3	8.1	8.6	10.9	12.8	15.2	17.7	(25)
(26)			MCWB	14.8	15.1	13.6	11.7	9.9	7.7	6.3	6.6	8.3	9.7	11.3	13.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.9	18.7	17.1	15.4	13.5	11.2	9.7	10.0	11.5	12.7	15.1	17.0	(27)
(28)			MCDWB	22.6	22.3	19.7	17.1	15.3	12.8	11.1	11.6	13.8	15.9	19.1	20.3	(28)
(29)		2%	WB	17.5	17.4	15.7	13.9	12.2	9.8	8.5	8.5	10.4	11.6	13.9	15.7	(29)
(30)			MCDWB	21.0	20.6	18.3	15.8	13.9	11.2	9.9	10.2	12.5	14.3	17.1	18.7	(30)
(31)		5%	WB	16.4	16.5	14.7	13.0	11.1	8.7	7.4	7.5	9.5	10.7	12.7	14.8	(31)
(32)			MCDWB	19.9	19.7	17.6	15.0	12.9	10.0	8.8	9.2	11.6	13.3	15.7	17.8	(32)
(33)		10%	WB	15.5	15.7	13.9	12.0	10.0	7.8	6.5	6.7	8.6	9.9	11.8	14.1	(33)
(34)			MCDWB	18.8	18.8	17.0	14.1	11.7	9.1	7.9	8.4	10.6	12.4	14.7	17.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.2	9.9	9.6	8.6	7.2	6.5	6.3	6.9	7.9	8.4	9.2	9.1	(35)
(36)			MCDBR	12.2	11.6	11.5	10.0	8.3	7.0	7.1	8.5	9.5	10.3	11.5	10.7	(36)
(37)			MCWBR	7.0	6.4	6.7	6.6	6.0	5.4	5.3	6.0	6.6	6.9	7.4	6.5	(37)
(38)		5% WB	MCDWR	10.0	9.4	9.0	7.7	7.0	6.0	6.0	7.0	8.0	9.1	10.0	9.1	(38)
(39)			MCWBR	6.3	5.6	5.7	5.5	5.3	5.0	4.9	5.4	6.0	6.6	6.9	6.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.310	0.306	0.289	0.281	0.266	0.264	0.260	0.270	0.291	0.303	0.298	0.310	(40)	
(41)		taud	2.577	2.604	2.663	2.664	2.700	2.675	2.694	2.664	2.566	2.538	2.580	2.578	(41)	
(42)		Ebn at Noon	1022	1002	982	927	885	855	884	928	961	994	1028	1027	(42)	
(43)		Edh at Noon	104	96	83	72	59	56	59	71	90	102	103	105	(43)	
(44)	All-Sky Solar Radiation	RadAvg	6.50	5.65	4.51	3.10	2.07	1.60	1.85	2.56	3.59	4.69	5.77	6.02	(44)	
(45)		RadStd	0.49	0.57	0.26	0.21	0.13	0.14	0.15	0.20	0.31	0.33	0.55	0.49	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	+0.74	+0.84	+0.64	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)		N/A	N/A	+0.59	N/A	N/A	N/A	-25	N/A	N/A	N/A

Nomenclature: See separate page